Syllabus

MATH 2063, Section 003 – Multivariable Calculus *Summer 2025*

Instructor Name: Ethan Davis
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Office Location: French Hall West, Room 4513
Office Hours (on Zoom): Tu, Wed, Th 12:00pm – 12:30pm, or by appointment.

***If you need to contact me**, email is my preferred method of communication. I do my best to respond to emails within 24 hours on a weekday and 48 hours on the weekend. Sometimes I miss an email or forget, so if more than 48 hours have passed, please send a follow-up!*

Course Overview

Math 2063 will include the study of lines and planes, vector-valued functions, partial derivatives and their applications, multiple integrals, and calculus of vector fields. Or equivalently, chapters 12 – 16 in the textbook.

Textbook: *Calculus: Early Transcendentals*. 9th Edition, by James Stewart, with WebAssign.

Pre-requisites: To take this course you must have taken the following course, MATH1062, min grade C-.

Learning Outcomes

Successful completion of the course means a student should be able to…

1. Graph and find the equations of a line, plane, and surfaces in space.
2. Find partial derivatives, directional derivatives, gradients and differentials of functions of several variables and use them to solve and interpret solutions of applied problems.
3. Find extrema of functions of several variables using the second partials test and Lagrange multipliers, and solve applied problems.
4. Evaluate iterated integrals and use them to find the area of plane regions.
5. Evaluate multiple integrals in rectangular, polar, cylindrical and spherical coordinates and use them to solve applications involving volume, surface area, density, moment and centroids.
6. Use Green's Theorem, the Divergence (Gauss's) Theorem and Stokes's Theorem.

Course Format

This section is an asynchronous online class. As this course is four credit hours, **you are expected to spend at least four hours a week on this course**. You are responsible for watching the lecture videos for each section as well completing the posted assignments on time. There will be homework due on MTW, weekly quizzes due on Fridays, as well as two exams and a final. Success in an asynchronous online course is highly contingent on your ability to plan and manage your workload effectively. **Please refer to the course calendar for the recommended pace of course material.**

You should expect a weekly announcement on Monday outlining the focus of the week.

Technology

As this is an asynchronous online course, **reliable and consistent internet access is required**. The weekly **quizzes require file uploading** of hand-written work. You can use pdf scanner apps such as Adobe Scan, Google Drive, CamScanner, etc. Lastly, exams will use HonorLock, so you will need **access to a computer with a webcam**. No calculators of any kind may be used on any assessments.

Quizzes and Exams

There will be weekly quizzes to be completed on Friday each week. The lowest quiz grade will be dropped. There are no make-up quizzes, so a missed quiz will be counted as a 0 unless a valid reason is presented to the instructor prior to the exam date or, if circumstances mandate, immediately afterward. Quizzes are timed and given on canvas. They require hand-written work and file uploading.

There will be two exams and a final exam. A missed exam will count as a 0 unless a valid reason is presented to the instructor prior to the exam date or, if circumstances mandate, immediately afterward.

Exam 1 (Chapters 12 – 14)	Friday, June 13th
Exam 2 (Chapter 15)	Friday, July 7th
Final Exam (Cumulative)	Friday, August 8th

Exams will be proctored using Honorlock. For more information on Honorlock, please refer to the Honorlock page on canvas.

You are **not** allowed to work with other students on quizzes or exams. Using external resources like Chegg, Stack Exchange, ChatGPT, etc. on assessments is considered academic misconduct.

Calculator Policy: No calculators of any kind may be used on quizzes and exams.

Homework

Homework sets will be electronically assigned and graded through Web Assign. There will be homework due on Monday, Tuesday, and Wednesday of each week (with a few exceptions). You are welcome to work with other students on homework. **Late Policy**: *you are allowed to make one 7-day extension request per homework on Web Assign.*

Grading

Your grade for this course is based on the following categories:

Quizzes	20%
Homework	30%
Exams	30% (2 exams, 15% each)
Final	20%

I will do my best to have the Friday quizzes graded by the following Wednesday. Exams may take a few days longer.

Drop Period Ends May 26 @ 11:59 pm EDT (*100% Refund and class removed from academic record)*

Last day to Withdrawal Ends August 1 @ 11:59 pm EST (*No Refund and class remains on academic record as W)*

Academic Integrity

The University Rules, including the Student Code of Conduct and other documented policies of the department, college and university related to academic integrity including the assessment policies listed above will be enforced. https://www.uc.edu/campus-life/conduct/student-code-of-conduct.html

The sanction for Academic Misconduct will be failure of the course.

(AI) Artificial Intelligence/Apps/Extensions

You should not use artificial intelligence engines (e.g. ChatGPT), cell phone apps (e.g. Symbolab, Photomath), or browser extensions (e.g. Transcript, Coursology) to produce work for this class. While these might help with checking your solutions, if misused they often provide incorrect and incomplete information. You might trick yourself into believing your calculus ability is stronger than it really is or worse—confuse yourself by introducing advanced topics beyond the scope of this course. Besides, the majority of course credit comes from in-person assessments where these resources are strictly prohibited. Use of AI on the Quizzes, Homework, Exams, or Final Exam will be reported as Academic Misconduct.

The sanction for Academic Misconduct will be failure of the course.

If you are in doubt as to whether you are using an online learning support appropriately, I encourage you to meet with me for advice.

Please make note of and review the College of Arts and Science's Two-Strikes Policy. We take these violations seriously https://www.artsci.uc.edu/information/undergrad/academic-policies.html.

Accessibility Policy

Students needing special considerations should meet with me as soon as possible to arrange reasonable provisions to ensure an equitable opportunity to meet all the requirements of this course. Some accommodations may require prior approval by the Accessibility Resources Office.

The University of Cincinnati is committed to providing all students with equal access to learning opportunities. Accessibility Resources is the official campus office that works to arrange for reasonable accommodations for students with an identified physical, psychological or cognitive disability (learning, ADD/ADHD, psychological, visual, hearing, physical, cognitive, medical condition, etc.) Students are encouraged to contact the Accessibility Resources Office to arrange for a confidential meeting to discuss services and accommodations. Contact should be initiated as soon as possible to allow adequate time for accommodations to be arranged. Additional information can be found here: https://www.uc.edu/campus-life/accessibility-resources.html

Student Religious Accommodations

Ohio law and the University's Student Religious Accommodations for Courses Policy 1.3.7 permits a student, upon request, to be absent for reasons of faith or religious or spiritual belief system or participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization and/or to receive alternative accommodations with regard to examinations and other course requirements due to an absence permitted for the above-described reasons. Not later than fourteen days after the first day of instruction in the course, a student should provide the instructor with written notice of the specific dates for which the student requests alternative accommodations. For additional information about this policy, please contact the Executive Director of the Office of Equal Opportunity and Access at (513) 556-5503 or oeohelp@UCMAIL.UC.EDU.

Communication of Changes

The professor reserves the right to make changes to the syllabus, including assignment due dates and test dates (excluding any officially scheduled examinations) when necessary to meet learning objectives or when unforeseen circumstances occur. These changes will be announced as early as possible so students can adjust their schedules. Be assured that I will communicate to you any changes to our schedule, syllabus or policies quickly and efficiently through **Canvas.** It is your responsibility to make sure Canvas has a valid email address and it is recommended that you check this email daily. All course documents will be available via link in Canvas as well.